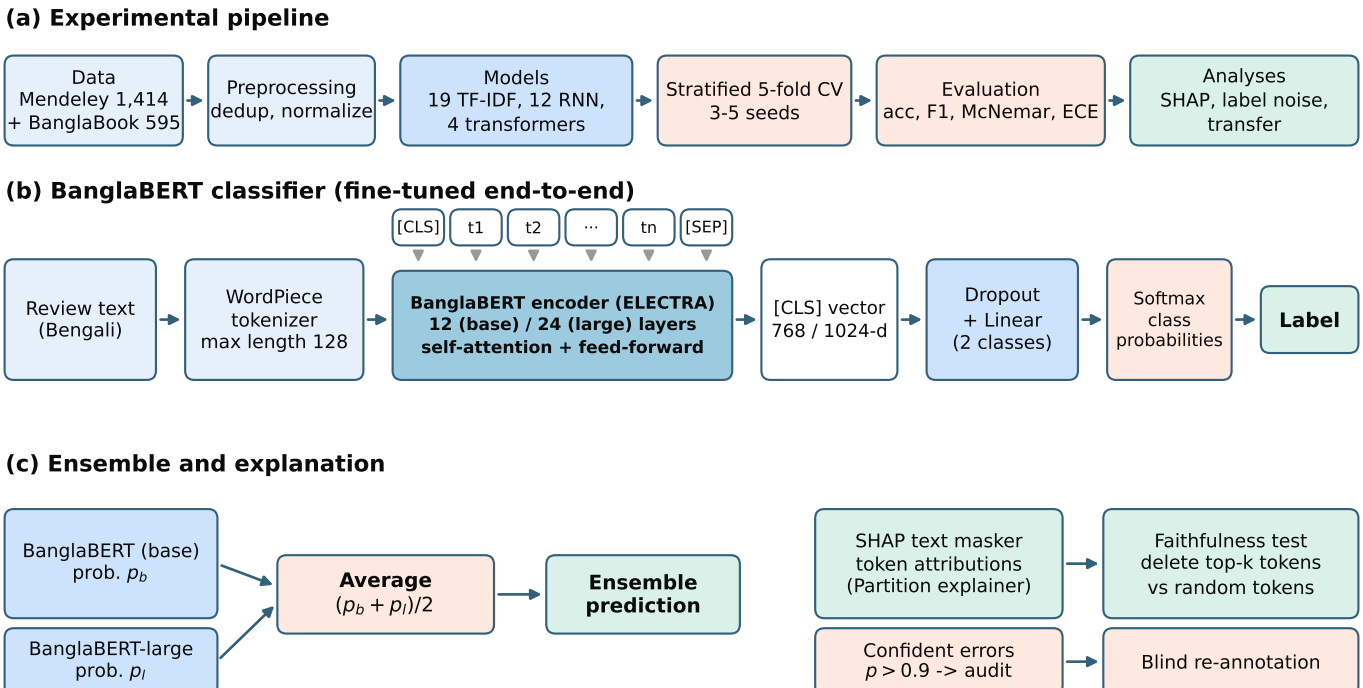
(a) Experimental pipeline
Data
Mendeley 1,414
+ BanglaBook 595
Preprocessing
dedup, normalize
Models
19 TF-IDF, 12 RNN,
4 transformers
Stratified 5-fold CV
3-5 seeds
Evaluation
acc, F1, McNemar, ECE
Analyses
SHAP, label noise,
transfer
(b) BanglaBERT classifier (fine-tuned end-to-end)
[CLS]
t1
t2
···
tn
[SEP]
Review text
(Bengali)
WordPiece
tokenizer
max length 128
BanglaBERT encoder (ELECTRA)
12 (base) / 24 (large) layers
self-attention + feed-forward
[CLS] vector
768 / 1024-d
Dropout
+ Linear
(2 classes)
Softmax
class
probabilities
Label
(c) Ensemble and explanation
BanglaBERT (base)
prob. p_b
BanglaBERT-large
prob. p_l
Average
$(p_b + p_l)/2$
Ensemble
prediction
SHAP text masker
token attributions
(Partition explainer)
Faithfulness test
delete top-k tokens
vs random tokens
Confident errors
$p > 0.9$ -> audit
Blind re-annotation